Empirical Study of Memorization in Differentially Privately Fine-Tuned Large Language Models

Yuwei (Johnny) Meng

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Link to GitHub Repo

Abstract

1. Introduction

In 2017, a group of researchers at Google Brain proudly announced the architecture of the transformer neural model, which revolutionized the field of Natural Language Processing (NLP) and led to the rise of large language models (LLMs). Training LLMs requires large amounts of data, and if not curated carefully, these datasets might contain sensitive information such as social security numbers or medical records. This poses a severe problem in model security and ethics, as shown by Carlini et al. (2019) that neural networks have a high risk of unintentionally memorizing fine-grained information and oddly-specific details. To mitigate this issue and protect the privacy of individuals, one approach is to leverage methods from *differential privacy* (DP), which provides a rigorous mathematical framework for quantifying privacy (Dwork et al., 2006). With DP, researchers have developed DP-based training methods, such as DP-SGD, to train neural networks and showed that they effectively mitigate privacy concerns during training. Background information about DP and DP-SGD utilized in this paper is described in Section 2.

Despite the prominent performance of DP-augmented training methods, the use of these methods in training or fine-tuning LLMs still remains underexplored in both research and industrial settings. Specifically, the benefits of DP in reducing unintentional memorization of training data are to be precisely quantified. Therefore, this study aims to narrow down this gap by investigating the following research question:

> *At what privacy budget ε does DP-SGD effectively prevent memorization of sensitive sequences in fine-tuned LLMs, and what is the accuracy cost of that protection?*

2. Background

Differential privacy is a mathematical framework that leverages probability theories to rigorously quantify privacy in algorithms (Dwork et al., 2006). Formally, let $\mathcal{X}$ denote the data space and $X = (x_1, \cdots, x_n) \in \mathcal{X}^n$ be a dataset. We can then define another dataset $X' = (x'_1, \cdots, x'_n) \in \mathcal{X}^n$ to be a *neighboring dataset* of X, denoted as $X \sim X'$, if there exists $i \in \{1, \cdots, n\}$ such that $x_i \neq x_i'$, and for all $j \neq i$, $x_j = x_j'$. With this setup, we say that a randomized algorithm $\mathcal{M}$ is (ε, δ)-*differentially private* if for all datasets $X \sim X'$, and all $S \subseteq \Omega$ where Ω is the output space of $\mathcal{M}$,

$$\mathbb{P}(\mathcal{M}(X) \in S) \leq e^{\varepsilon}\mathbb{P}(\mathcal{M}(X') \in S) + \delta.$$

In the context of training deep neural networks, we can adapt this definition and make gradient descent a differentially private algorithm so that the trained models do not memorize specific details of any training data. Defining $\mathcal{L}$ as the loss function, α as the learning rate, instead of updating the weights by $\alpha \nabla \mathcal{L}$ at every step, we first compute each per-sample gradient $\nabla \mathcal{L}_i$ and clip it to the L2 norm bound C, then sum the clipped per-sample gradients and add random noise from $\mathcal{N}(0, \sigma^2 C^2 \boldsymbol{I})$ to the sum before updating. By choosing the parameters σ and C strategically, one can show that this algorithm, named *differentially private stochastic gradient descent* (DP-SGD), satisfies (ε, δ)-differential privacy (Abadi et al., 2016).

3. Related Work

4. Methodology

5. Results & Discussion

6. Conclusion

6.1. Limitations & Future Directions

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